Covering 1024 syndromes with 50 columns

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with instructions from Stephen Wu

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Abstract

A binary linear $[50, 40]_2$ code of covering radius 2 certifies $\ell_2(10, 2) \leq 50$, one column below the Kaikkonen–Rosendahl length 51 that has stood since 2003 and that still seeds the $R = 2$ family of Davydov–Marcugini–Pambianco (arXiv:2511.02542). The new matrix admits a $(2, 0)$-partition with $p(\boldsymbol{H}) = 10$, so Construction QM_2^2 propagates it to exhaustively verified codes of lengths 815 and 1631 at $r = 18$ and $r = 20$, and to the family $n = 51 \cdot 2^{r/2-5} - 1$ of asymptotic density 2601/2048. This note explains the problem, how the matrix was found, what was proved, and how to re-check every number from the committed text. The matrices, verifiers, and LaTeX source live at `https://github.com/wustep/maths`.

Contents

1 The problem

1.1 Covering radius

Let $C \leq \mathbb{F}_2^n$ be a binary linear code of codimension r, so $|C| = 2^{n-r}$. Its *covering radius* is the least R such that every vector of $\mathbb{F}_2^n$ lies at Hamming distance at most R from some codeword. Equivalently, if $\boldsymbol{H}$ is an $r \times n$ parity-check matrix with column set $S \subset \mathbb{F}_2^r$, every syndrome is a sum of at most R columns of $\boldsymbol{H}$. Write $\ell_2(r, R)$ for the least such length.

A binary linear code of length n and dimension $k = n-r$ is written $[n, k]_2$; when the covering radius is R we append it, $[n, k]_2\, R$. So a $[50, 40]_2\, 2$ code is binary of length 50 and dimension 40, has covering radius 2, and hence redundancy $r = 10$.

For $R = 2$ the condition is

$$\{0\} \cup S \cup (S + S) = \mathbb{F}_2^r. \tag{1}$$

A $(2, 0)$-partition of the columns (Definition 3.2 of [1]) is a partition into nonempty blocks such that every syndrome, including 0, is a sum of at most two columns from *distinct* blocks. Write $p(\boldsymbol{H})$ for the number of blocks.

The finite covering density of an $[n, n-r]_2 2$ code is the exact rational

$$\mu = \frac{1 + n + \binom{n}{2}}{2^r}.$$

Green's Problem 40 asks for the asymptotic density

$$f(2) = \liminf_{r\to\infty} \mu\big(\ell_2(r, 2), r\big).$$

For an infinite family of $[n(r), n(r) - r]_2\, 2$ codes, write $\bar{\mu}(2)$ for the $\liminf$ of μ along the family as $r \to \infty$. Green's $f(2)$ takes the $\liminf$ over the *optimal* lengths $\ell_2(r, 2)$, so any family bound $\bar{\mu}(2) \leq c$ gives $f(2) \leq c$. It is not even known whether $f(2) = 1$. The pre-2025 upper bound $f(2) \leq 729/512 \approx 1.42383$ had stood since 1992. Davydov, Marcugini and Pambianco [1] brought it to $2704/2048 \approx 1.32031$ in November 2025, by iterating a construction they call QM_2^2 from a single seed. QM_2^2 grows a short radius-2 covering code into a longer one, provided the seed admits a $(2, 0)$-partition into few enough blocks; its exact input–output formula is recalled in Section 3.5.

1.2 Why the $r = 10$ cell is load-bearing

Their Table 5.1, row $r = 10$, still lists $n = 51$: the Kaikkonen–Rosendahl [2] code of 2003, density $1327/1024 \approx 1.29590$. The entry is not bolded. Even the 2025 paper did not improve it.

Their infinite family (Theorem 5.7) has lengths

$$\Phi(r) = 26 \cdot 2^{r/2-4} - 1 = 2^m(51 + 1) - 1, \qquad m = r/2 - 5.$$

The 52 is the Kaikkonen–Rosendahl 51, plus one. The whole family is that one table cell pushed forward. Theorem 5.7(i) says the 2-partitions of $\boldsymbol{H}_{KR}$ are "very important for the next iterative process".

So a 50-column matrix at $r = 10$ is not a one-cell footnote. If it admits a small enough $(2, 0)$-partition, it re-seeds the family. The paper's computer search found $p(\boldsymbol{H}_{KR}) = 11$.

2 The process

The searcher that found the matrix was run overnight on 16 August 2026 by an automated agent (Codex `gpt-5.6-sol`, Max). A committed, deterministic reproduction of that run, `compute/search_radius2.py`, encodes every constant quoted below; none of this section is from memory.

2.1 Encoding and seed

Every matrix in this folder stores a column as an unsigned integer whose bit i, counting from the least significant bit, is row $i+1$. The ten identity columns are $1, 2, 4, \ldots, 512$.

The seed was the Kaikkonen–Rosendahl 51-set: the ten identity columns plus the 41 hex words printed in Theorem 4.3 of [1]. The searcher reads those hex words directly as LSB-first integers. The paper prints them MSB-first, so the matrix the searcher actually holds is $\boldsymbol{H}_{KR}$ with its rows in reverse order. Row reversal is an invertible linear map on $\mathbb{F}_2^{10}$, so the reversed matrix is still a rank-10 covering matrix of radius 2, and no downstream claim depends on the orientation.

The 50-subset was formed by trying all 51 single deletions: removing `0x1A3` $= 419$ leaves 11 of the 1024 syndromes uncovered, and no deletion does better. The search therefore starts eleven holes away from a witness.

2.2 State, energy, and the incremental update

The state is a set of exactly 50 distinct nonzero ten-bit columns; the cardinality never changes. An array of 1024 counters records, for every syndrome, how many of the $1 + 50 + \binom{50}{2} = 1276$ representations hit it: the empty sum, the 50 singletons, and the unordered pair-XORs. The energy is the number of counters at zero; a witness is a state of energy 0.

Swapping one column is cheap. Removing column c decrements its own counter and the 49 counters $c \oplus c'$ for the columns c' that stay; inserting the replacement increments the same pattern. That is about 100 counter updates per proposal instead of a 1276-term recount, and a rejected proposal is undone the same way.

2.3 Proposals, temperature, and restarts

With probability 220/256 a proposal aims at a hole: pick an uncovered syndrome g uniformly, then offer the column g itself with probability 1/64, otherwise $g \oplus s$ for a uniformly chosen current column s. The first offer covers g as a new singleton; the second covers it as a pair through s, unless the compensating deletion happens to remove s. The searcher draws up to 100 times for an offer that is nonzero and not already a column; failing that, and in the remaining 36/256 of proposals, the offer is a uniform random non-member. The column to delete is uniform among the 50.

Acceptance is Metropolis on the change in energy: downhill and level moves always pass, uphill moves pass with probability $e^{-\Delta/T}$. The temperature runs in cycles of 80,000 proposals; within a cycle, $T = 3.5 \cdot 0.015^{\phi}$ as ϕ goes from 0 to 1, so each cycle cools from 3.5 to about 0.05. At the end of a cycle the state resets to the best configuration seen so far.

The random source is xorshift64 with shift constants 13, 7, 17 and a fixed seed constant; that is run 0. Run 0 reached energy 0 at proposal 3,600,281, which is 281 proposals after its 45th restart from best. The witness keeps 35 of the 50 starting columns; the anneal replaced 15.

Two replay commands. `python3 compute/search_radius2.py` re-checks the committed witness using the standard library alone. `python3 compute/search_radius2.py --search` replays discovery run 0 deterministically; it needs NumPy and Numba.

2.4 What the same search did not close

The identical machinery failed on the prettier one-column gaps. At $(r, n) = (8, 25)$ the best anneal missed 3 of 256 syndromes; CP-SAT returned `UNKNOWN` after 240 seconds and Z3 after 180. At $(9, 38)$, CP-SAT returned `UNKNOWN` after 300 seconds, and every single deletion from a reconstructed Gabidulin 39-set missed at least 8 of 512. At $n = 49$, $r = 10$, the anneal stalled at 7 of 1024. All of these are search residues, not lower bounds.

Later sessions measured how deep the $n = 49$ residue is (`ATTACK.md`, 2026-08-18 and 19). Fresh deterministic anneals stall at exactly 7 holes again. The best 7-hole configuration admits no swap of five or fewer columns reaching zero holes, an exhaustive claim over all $\binom{49}{5} = 1{,}906{,}884$ removal sets with an exact re-add search, validated on four out of four planted controls. A second 7-hole optimum admits no swap of four or fewer columns, and a color-guided backtracking search proves that no linear map of $\mathbb{F}_2^{10}$ carries one 49-set to the other, so the landscape has at least two distinct deep basins at 7 holes. Separately, 79 subgroup classes of $GL(10, 2)$ were exhausted with zero invariant 49-coverings. None of this bounds $\ell_2(10, 2)$ from below; it says the known near-misses are far from any witness. Replay: `python3 compute/q4/verify_config.py compute/q4/best_sa_config.cols` prints the seven holes, and the exhaustion logs are `compute/q4/kswap5_sa.log` and `compute/q4/orbit_runs.log`.

2.5 Where the partition came from

The overnight run stopped at the table cell; it never computed a partition. The 10-block $(2, 0)$-partition was produced the same day in an intermediate session, and it entered the repository as an unverified claim: the brief handed to the verification session listed the ten blocks and instructed that session to check them from scratch and stop if any check failed. The intermediate session's scripts were deleted under a quarantine protocol once verification passed, so that the two checkers would remain the only implementations in the tree. The code that first found the partition is therefore not preserved. What is committed is the partition itself, `result/data/partition_p10.json`, consulted for block labels only, and two independent checkers that re-derive the columns from the matrix text and sweep all 1024 syndromes.

The certified pair statistics show what any such search is up against. Of the 973 syndromes that need a pair, 821 have a unique one, and each unique pair forces its two columns into distinct blocks. Those constraints form a graph on the 50 columns with 821 edges, and a valid $(2, 0)$-partition is a proper coloring of that graph which, for each of the remaining 152 syndromes, also splits at least one of its 2 to 6 pairs across blocks. So most of the partition is forced before any search begins, and checking a candidate costs one pass over the 1276 sums. No claim is made that 10 blocks is the minimum.

Ten blocks is what the propagation needs, and less than what was published. Construction QM_2^2 requires $p(\boldsymbol{H}_0) \leq 2^m \leq n_0$, so $p = 10$ keeps both $m = 4$ and $m = 5$ legal with room to spare, and it beats the computer-searched $p(\boldsymbol{H}_{KR}) = 11$ of [1] on a matrix one column shorter.

2.6 From search output to artifact

The overnight folder never ran QM_2^2 and still quoted the stale $f(2) \leq 1.4238$.

A later session (Claude, afternoon of the same day) treated everything upstream as a claim to be verified, not as a trusted input: reconstructed the Kaikkonen–Rosendahl baseline from the hex in Theorem 4.3 of [1], checked the 50-set and the partition, implemented QM_2^2 from equations (4.2) and (4.4), and exhaustively checked the $r = 18$ and $r = 20$ outputs. Two verifiers were written from scratch in two languages, then a third quarantined oracle was run once as a post-hoc diff. The search procedure is not part of the argument. The argument is a finite decidable statement about committed matrices, re-checkable in seconds.

3 The result

3.1 The code

claim	value
shape	10×50, $\mathbb{F}_2$-rank 10, 50 distinct nonzero columns
coverage	1024/1024 syndromes
covering radius	exactly 2
density μ	$319/256 = 1.24609375$
minimum distance	$d = 3$
KR baseline	51 columns, 1024/1024, density 1327/1024

The columns, as unsigned integers with bit i (LSB first) equal to row $i + 1$:

1	2	4	15	16	32	65	86	128	173
183	202	212	247	256	297	320	329	341	366
373	381	391	403	438	460	479	491	502	559
576	608	653	734	742	754	771	777	789	821
846	855	869	881	893	897	927	981	1003	1004

The multiplicity histogram of the $1276 = 1 + 50 + \binom{50}{2}$ representations over the 1024 syndromes is $\{1{:}859,\ 2{:}129,\ 4{:}24,\ 5{:}9,\ 6{:}3\}$. There are no zeroes. Exactly 973 syndromes are neither 0 nor a column, so the radius is not 1.

3.2 Minimality

Deleting any one of the 50 columns leaves at least 9 syndromes uncovered. The best deletions, columns 381, 479 and 927, leave exactly 9. So the 50-set is a minimal 1-saturating set in $PG(9, 2)$, equivalently a locally optimal (LO) covering code (the geometric name for a covering-radius-2 column set with no redundant column). It is not padded.

3.3 A $(2, 0)$-partition with $p(\boldsymbol{H}) = 10$

block	columns
0	$2, 128, 202, 212, 771, 855, 897, 981$
1	86
2	$381, 893, 1003$
3	$183, 297$
4	$1, 65, 247, 256, 320, 438, 502, 734$
5	$15, 173, 329, 366, 460, 559, 653, 846$
6	$4, 16, 391, 491, 742, 754, 869, 881$
7	$479, 1004$
8	403
9	$32, 341, 373, 576, 608, 777, 789, 821, 927$

All 973 pair-needed syndromes have a cross-block pair. The pair-only histogram over those 973 is $\{1{:}821,\ 2{:}123,\ 4{:}19,\ 5{:}8,\ 6{:}2\}$, so 821 syndromes have a unique pair and force their two columns into distinct blocks. Against the paper's $p(\boldsymbol{H}_{KR}) = 11$.

3.4 The dependent triple

There are exactly 10 linearly dependent triples of columns. Exactly one, $(491, 734, 821)$, lies in three distinct blocks (blocks 6, 4, 9). This is the analogue of [1] Theorem 5.2(ii), the ingredient Construction QM_5^2 needs at $r = 28$. That construction is not part of `result/`; a later session ran it (Section 3.7).

3.5 Propagation

Construction QM_2^2 ([1], Theorem 4.1) takes an $[n_0, n_0 - r_0]_2 2$ code with a 2-partition into $p(\boldsymbol{H}_0)$ blocks, an indicator set $\mathscr{B} = \mathbb{F}_{2^m}$ subject to $n_0 \geq 2^m \geq p(\boldsymbol{H}_0)$, and $\boldsymbol{D} = \boldsymbol{D}_1(2)$, and returns

$$n = 2^m(n_0 + 1) - 1, \qquad r = r_0 + 2m, \qquad p(\boldsymbol{H}_C) \leq 2^{m+1} + 1.$$

With $n_0 = 50$ and $p(\boldsymbol{H}_0) = 10$ the inequality permits exactly $m = 4$ and $m = 5$ ($2^6 = 64 > 50$). Both outputs were enumerated completely:

	r	n	coverage	density	published
seed	10	50	1024/1024	319/256	51 (KR 2003)
$m = 4$	18	815	262144/262144	332521/262144	831
$m = 5$	20	1631	1048576/1048576	1330897/1048576	1663

A second pair of matrices, from a different indicator allocation, also verifies (see `result/data/alt/`). Indicator assignment is a free choice, so two allocators give two codes.

3.6 The family

Iterating from $(r, p) = (10, 10)$ with the paper's bound $p(\boldsymbol{H}_C) \leq 2^{m+1} + 1$, the even $r \leq 64$ reachable by QM_2^2 are

$$10, 18, 20, 30, 32, 34, 36, 38, 40, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64.$$

The even r *not* reachable this way are $12, 14, 16, 22, 24, 26, 28, 42, 44$. Those gaps are real. The paper fills its own $r = 22, 24, 26$ with QM_3^2 and $r = 28$ with QM_5^2; the analogues are not implemented here.

Closed form on the reachable even r:

$$n = 51 \cdot 2^{r/2-5} - 1, \qquad \mu = \frac{51^2}{2^{11}} - \frac{51}{2^{r/2+1}} + \frac{1}{2^r} \nearrow \frac{2601}{2048} \approx 1.27002.$$

Hence $\bar{\mu}(2) \leq 2601/2048$, and since $f(2)$ is a $\liminf$ over r, also $f(2) \leq 2601/2048$. Against the literature:

	$\bar{\mu}(2)$	source
pre-2025	$729/512 \approx 1.42383$	standing since 1992
arXiv:2511.02542	$2704/2048 \approx 1.32031$	seeded by KR $n_0 = 51$
this seed	$2601/2048 \approx 1.27002$	seeded by $n_0 = 50$

Rows past $m = 5$ inherit the paper's $p(\boldsymbol{H}_C)$ bound rather than computed partitions. That is the weakest link.

3.7 Later confirmations of the paper's constructions

After this note was first written, further sessions implemented other constructions of [1] on the same seed: QM_3^2, QM_5^2 and QM_2^1 at radius 2, and radius-3 and radius-4 variants. Each was built from the paper text and checked by an independent matrix-only verifier; the radius-2 outputs at $r = 22, 24, 26, 28$ were swept exhaustively, up to all 2^{28} syndromes at $r = 28$. Those runs confirm that the paper's constructions operate as stated from the 50-column seed. The certified values and replay scripts are recorded in `ATTACK.md` and `compute/`; this note stays with the seed, the partition, and the $m = 4, 5$ lifts.

4 Verification

4.1 How to re-run

The code is the public repository `https://github.com/wustep/maths`. Clone it, then run the pipeline from the covering-result directory:

```
git clone https://github.com/wustep/maths.git
cd maths/problems/covering/result
./run_all.sh
```

Needs `python3` and `rustc`. After the clone, no network, no packages, no RNG. Seventy-four named assertions, exit 0, output byte-identical across runs (about two seconds). Direct links: `https://github.com/wustep/maths/tree/main/problems/covering/result` and `result/run_all.sh`.

4.2 Two verifiers, opposite algorithms

`verify/verify.py` is pair-driven: walk all $\binom{n}{2}$ pairs, mark the syndromes they hit, then sweep 2^r for anything unmarked. `verify/verify.rs` is syndrome-driven: for each of the 2^r syndromes, scan the columns and test membership of $s \oplus h$. It then runs its own pair loop and refuses to report unless the two verdicts agree. They also differ in parser, rank pivot (lowest vs highest set bit), and exact rational arithmetic. `run_all.sh` diffs their fact dumps.

Both read the matrix from *text*. Neither consults a stored certificate. The partition JSON is used only for block labels; the columns themselves are re-derived from the matrix.

4.3 Encoding

Everything in the repo is LSB-first: bit i of a column integer is row $i + 1$. The hex listing of M_{KR} in [1] Theorem 4.3 is the opposite (MSB-first, row 1 as the high bit). Reconstructing $\boldsymbol{H}_{KR} = [I_{10} \mid M_{KR}]$ reverses the ten bits. A from-scratch verifier that misses the reversal fails on the Kaikkonen–Rosendahl baseline *and only there*, which is a useful signal. The reversal is pinned by the paper's own Theorem 5.2(ii), $h_5 + h_{27} + h_{29} = 0$, asserted in the builder as a canary.

A reader using the opposite convention gets the bit-reverse of every column, an $\mathbb{F}_2$-linear relabelling of $\mathbb{F}_2^{10}$, and every claim here is preserved.

4.4 Independence, honestly

Both verifiers were written in one session by one author. Two languages, two algorithms, two parsers, no shared code, but not two people. A third, quarantined oracle was run once after both were green and agreed everywhere; it is not in the tree.

5 What is not claimed

- 50 is not shown optimal. Sphere-covering gives only $\ell_2(10,2) \geq 45$, since $1+n+\binom{n}{2} \geq 1024$ fails at $n = 44$, which gives only 991. Minimality says this particular 50-set has no redundant column, which says nothing about a different 49-set.
- The $n = 49$ runs leaving 7 uncovered syndromes are search residue, not a lower bound. The residue is deep (no $\leq$ 5-column swap fixes the best near-miss, and there are two inequivalent near-misses; see Section 2.4), but depth of a residue is still not a bound.
- The gaps $(8, 25)$ and $(9, 38)$ remain open.
- Only $m = 4$ and $m = 5$ are exhaustively verified. The rest of the family inherits $p(\boldsymbol{H}_C) \leq 2^{m+1} + 1$ from [1] (4.4).
- Within `result/`, nothing is claimed for $r \in \{12, 14, 16, 22, 24, 26, 28, 42, 44\}$. Later work recorded in `ATTACK.md` implemented QM_3^2 and QM_5^2 from this seed and certified $r = 22, 24, 26, 28$ (Section 3.7); $r = 12, 14, 16$ still carry no claim from this seed.
- $f(2) \leq 2601/2048$ is an upper bound only. Whether $f(2) = 1$ is untouched.
- Priority is unresolved. Table 5.1 of [1] is "best as far as the authors know". A $\ell_2(10,2) \leq 50$ may already exist in ACCT proceedings. See `result/PRIORITY.md` in the repository.

6 Where the files live

This document is self-contained. Everything named below is in the public repository `https://github.com/wustep/maths`, under `problems/covering/` on `main`.

in the repo	what
`https://github.com/wustep/maths`	the repository
`result/NOTE.md`	repo companion
`result/note.tex`	arXiv-style draft of the same claims
`result/run_all.sh`	the pipeline
`result/data/H_r10_n50.txt`	the matrix
`result/data/partition_p10.json`	the 10-block partition
`compute/search_radius2.py`	the searcher, deterministic replay
`explainer.html`	interactive companion
`WALKTHROUGH.md`	how the overnight search felt
`ATTACK.md`	chronological log

The HTML explainer is `explainer.html`. The matrix is `H_r10_n50.txt`.

A The matrix as bits

Rows 1 to 10 from top to bottom, columns left to right. Bit i (LSB) of each integer column is row $i + 1$.

```
10010010011001010110111100110100100011110111111110
01010001001101000001001110111100011110001100001010
00110001011011000011111011101100111000111110101101
00010000010100010101010001110100110001001000101011
```

```
00001001001011000010110110101000010100110101101100
00000100011001010001110010011101001100010011100011
00000011000111001111110001111011011100001111100111
00000000111111000000001111111000111100000000011111
00000000000000111111111111111000000011111111111111
00000000000000000000000000000111111111111111111111
```

References

[1] A. A. Davydov, S. Marcugini, F. Pambianco, *New upper bounds for binary linear covering codes*, arXiv:2511.02542 [cs.IT], November 2025.

[2] M. Kaikkonen, C. Rosendahl, *New covering codes from an ADS-like construction*, IEEE Trans. Inform. Theory **49** (2003), no. 7, 1809–1812.

[3] G. Cohen, I. Honkala, S. Litsyn, A. Lobstein, *Covering Codes*, North-Holland Mathematical Library **54**, Elsevier, Amsterdam, 1997.

[4] B. Green, *100 open problems*, Problem 40. `https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf`.